

FLUTTER MOBILE APP

PROJECT PLAN — SUBMISSION

Momentum

A clean, motivating habit tracker that turns small daily actions into lasting streaks.

PLATFORM

Flutter · Android / iOS

SUBMITTED

03 August 2026

FINAL DEADLINE

21 August 2026

A semester project demonstrating navigation, forms, validation, Provider state management, SharedPreferences persistence, and custom reusable widgets.

MOMENTUM

PROJECT PLAN

Project Plan

Momentum — a Flutter habit tracker application.

Project Title	Momentum	Platform	Flutter (Android / iOS)
Category	Productivity / Self-improvement	State Management	Provider + SharedPreferences
Plan Submitted	03 August 2026	Final Submission	21 August 2026

SECTION 01

Project Title

Momentum — a mobile habit-tracking application built with Flutter.

SECTION 02

Project Description

Momentum is a lightweight, offline-first Flutter application that helps users build and maintain positive daily habits. The core idea is simple: small actions, repeated consistently, compound into meaningful change. The app lets a user define the habits they want to track, mark them complete each day, and visualise their progress through streaks, completion rings, and weekly statistics.

The project is intentionally scoped to be fully deliverable within the semester timeline while still exercising every major concept taught in class — multi-screen navigation, forms with validation, the Provider pattern for state management, SharedPreferences for local persistence, and a library of custom, reusable widgets. Special attention is given to clean architecture: a clear separation between the UI layer, the state layer, and the data layer, so the codebase reads as a portfolio piece rather than a prototype.

SECTION 03

Problem Statement

Most people know which habits would improve their lives — drinking more water, reading, exercising, studying consistently — yet struggle to maintain them. The two most common reasons are **forgetting** and a **lack of visible progress**. Without a reliable reminder and without seeing how far a streak has

come, motivation fades within a couple of weeks.

Existing habit apps tend to fall into one of two extremes: either they are stripped so bare that they offer no real insight into progress, or they are overloaded with social features, subscriptions, and account requirements that get in the way of the single task a user opens the app for. Momentum addresses the gap in the middle — a focused, fast, offline-first tool whose only job is to help a user show up every day and see their consistency made visible.

SECTION 04

Target Users

Momentum is designed for anyone who wants to build consistency in their daily routine. The primary audience includes:

- **Students** who want to build study routines, track revision, or maintain sleep and exercise habits alongside their academic schedule.
- **Working professionals** looking to establish morning routines, focused work blocks, or wellness habits such as hydration and stretching.
- **Self-improvement beginners** who need a simple, low-friction entry point — no account, no sign-up, no overwhelm; just open the app and check off a habit.
- **Privacy-conscious users** who prefer a fully offline app that stores data only on their device rather than in the cloud.

SECTION 05

List of Core Features

Each feature below maps directly to one or more concepts from the course, so the app doubles as a practical demonstration of the semester’s syllabus.

Feature	What it does
Habit Management (CRUD)	Create, edit, archive, and delete habits with a name, icon, colour, and target frequency. Demonstrates forms, input validation, and custom dialogs.
Daily Check-in	A today-focused home screen where the user taps to mark each habit complete for the current day. Powers the streak engine in real time.
Streak Tracking	Automatically computes current and longest streaks per habit, with gentle handling of missed days so a single slip does not erase a month of progress.
Progress Visualisation	Completion rings for today, a 7-day weekly bar view, and a month heatmap. Built as custom painted widgets to showcase Flutter’s rendering.

Feature	What it does
Statistics Dashboard	Aggregated stats — total completions, average completion rate, best habit, and areas needing attention — powered by Provider-provided state.
Reminders & Notifications	Per-habit daily reminders scheduled through local notifications so the user is nudged without the app being open.
Light & Dark Theme	A user-toggable theme persisted via SharedPreferences, applied app-wide through a Provider-delivered theme controller.
Onboarding & App Settings	A first-launch onboarding flow plus a settings screen for time format, week start day, and data export/reset.
Offline-First Persistence	All habits and check-in history are stored locally with SharedPreferences (and a serialised model), so the app works fully offline.

Concept coverage. The table below shows how each course concept is exercised.

Course Concept	Where it appears in Momentum
Multi-screen Navigation	Home → Add/Edit Habit → Statistics → Settings, wired with named routes and a bottom nav bar.
Forms & Validation	Add/Edit Habit form validates name length, duplicate names, and frequency selection before saving.
Provider (State Management)	A HabitProvider exposes the habit list and check-in state; ThemeProvider and SettingsProvider handle preferences.
SharedPreferences	Persists habits, check-in history, theme choice, and onboarding completion across app restarts.
Custom Widgets	HabitCard, StreakRing, WeekBarChart, MonthHeatmap, and SectionHeader are built as reusable components.
Lists & Scroll Views	ListView/GridView for the habit list; slivers for the statistics dashboard.

SECTION 06

Development Timeline

A realistic, week-by-week plan from plan submission to final delivery.

Week	Dates (2026)	Milestone
1	03 Aug – 09 Aug	Project setup, folder architecture, theme, and navigation skeleton.
2	10 Aug – 16 Aug	Habit model, SharedPreferences persistence layer, and HabitProvider.

Week	Dates (2026)	Milestone
3	17 Aug – 21 Aug	Add/Edit forms with validation, daily check-in, and streak logic.
4	final by 21 Aug	Statistics dashboard, custom widgets polish, testing, and final submission.